COVID-19 Pandemic Analytics Data Warehouse

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Abstract— This paper presents the design and implementation of a data warehouse for global COVID-19 analytics using dimensional modeling techniques. The original flat, wide dataset from Our World in Data (OWID) was transformed into a Kimball-style star schema comprising one central fact table and seven-dimension tables. Full third normal form (3NF) was achieved, and the fact table was optimized for analytical queries. The solution deployed on Azure SQL Database, successfully handles sparse data attributes with high cardinality dimensions. The final model contains exactly 50,000 fact rows with 100% referential integrity and supports complex analytical queries across cases, deaths, testing, vaccination rollout, excess mortality, and reproduction rate.

I. Introduction

The COVID-19 pandemic generated one of the largest and most detailed public health datasets in history. The Our World in Data (OWID) COVID-19 dataset provides daily metrics for over 200 countries from January 2020 onward, including cases, deaths, testing, vaccination progress, hospital admissions, reproduction rate, and government stringency [1]. However, the raw dataset is wide, denormalized, and contains significant sparsity and repeating attribute groups, which makes it unsuitable for efficient analytical reporting.

In this project, I transformed the OWID dataset into a scalable data warehouse that is optimized for analytics. Kimball dimensional modeling principles was also used and implemented in Azure SQL Database.

II. Dataset Description

The dataset consists of daily observations per country with the following major metric categories:

- Total cases, new cases, cases per million
- Deaths and excess mortality
- Hospital and ICU admissions
- Testing volume and positivity rates
- Vaccinations administered, people vaccinated, boosters
- Country demographics like population, reproduction rate, median age, life expectancy, etc.

The extract contains approximately 50,000 rows covering the years 2020–2025.

III. Database Architecture and Design Process

I selected the Kimball dimensional modeling method because it is the model most widely used for analytical workloads. Key design decisions:

- All dimension tables normalized to at least 3NF
- Use of surrogate keys for all dimensions
- Dimensions with high cardinality with metric groups like cases, deaths, testing, vaccination, etc.

The final architecture is a star schema with one fact table and seven dimension tables.

IV. Schema Design and Relationships

TABLE I
STAR SCHEMA TABLES AND CARDINALITY

Table Name	Type	Primary Key	Approx. Rows	Description
Dim_Date	Dimension	date_id	2,192	Full calendar 2020–2025
Dim_Location	Dimension	location_id	~220	Countries with static demographics
Dim_Case	Degenerate Dimension	case_id	~15,000	Distinct case metric combinations
Dim_Death	Degenerate Dimension	death_id	~12,000	Death and excess mortality combinations
Dim_HospitalPatient	Degenerate Dimension	patient_id	~5,000	Hospital and ICU occupancy combinations
Dim_Vaccination	Degenerate Dimension	vaccinated_id	~8,000	Vaccination rollout combinations
Dim_Test	Degenerate Dimension	test_id	~6,000	Testing volume and positivity combinations
Fact_Daily_COVID	Fact Table	daily_id	50,000	One row per country per day

All relationships are enforced via foreign key constraints. A unique constraint on (date_id, location_id) in the fact table ensures exactly one row per country-date combination.

V. Implementation: DDL and ETL Process

Tables were created using standard DDL with surrogate integer/bigint keys and appropriate data types. The DDL script for table creation is provided in Listing 1.

Listing 1. DDL Script for Creating Tables

```
CREATE TABLE Dim_Date
(
    date_id INT IDENTITY(1,1) PRIMARY KEY,
    date DATE NOT NULL UNIQUE,
    year INT NOT NULL,
    month INT NOT NULL,
    day INT NOT NULL,
    month_name VARCHAR(20) NOT NULL,
    quarter INT NOT NULL,
    day_of_week INT NOT NULL,
    day_name VARCHAR(10) NOT NULL,
    is_weekend BIT NOT NULL
);
GO
```

The ETL process followed these steps:

1. Creation of a staging table matching the raw CSV structure
2. Population of Dim_Date using a recursive CTE
3. Population of Dim_Location with DISTINCT country attributes
4. Population of degenerate dimensions using DISTINCT metric combinations
5. Final fact table load using GROUP BY (date_id, location_id) and MAX(surrogate_key) to guarantee uniqueness and handle any residual duplication

The insertion scripts for populating the tables are provided in Listing 2.

Listing 2. DML Script for Inserting Data (ETL Population)

```
INSERT INTO Dim_Location
    (
    country, continent, population, population_density,
    median_age, life_expectancy
    )
SELECT DISTINCT
    country,
    continent,
    population,
    population_density,
    median_age,
    life_expectancy
FROM CovidProject
WHERE country IS NOT NULL
ORDER BY country;
GO
```

VI. Challenges and Solutions

Several data quality issues were encountered and resolved:

Challenge	Solution Applied
Duplicate (country, date) rows in staging	Used GROUP BY and removed duplicates using ROW_NUMBER() partitioning
UNIQUE constraint violations during load	Used TRUNCATE + GROUP BY + MAX() pattern to ensure single row
NULL key matching in metrics	Used ISNULL(column, 0) = ISNULL(column, 0) logic in joins

VII. Results and Analytical Capabilities

I made several analytical queries covering:

- Peak waves and case fatality rates
- Excess mortality rankings
- Testing positivity peaks
- Reproduction rate streak analysis
- Hospital/ICU occupancy peaks

VIII. Conclusion

I successfully transformed a complex, wide, and sparse COVID-19 dataset into a clean, scalable, and dimensional data warehouse using the Kimball methodology. I demonstrated full use of dimensional modeling, ETL design, referential integrity, and analytical query optimization.

References

[1] Our World in Data, “Coronavirus (COVID-19) Data,” 2025. [Online]. Available: https://ourworldindata.org/coronavirus